

LeetCode 2685: Count Complete Components

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Logic & Approach Notes

- basically dsu se krna he to
- $\text{edges} = (k * k - 1) / 2$
- k nikalne ke liye number of nodes chahiye in a component
- aur curren edge count chahiye
- node nikaalte waqt saare node consider krenge
- edge nikaalte waqt sirf u consider krenge jo edges me dekh rkha he

1. Disjoint Set Union (DSU) Implementation

```
class dsu{
public:
    vector<int> parent;
    vector<int> size;

    dsu(int n){
        parent.resize(n);
        size.resize(n,1);
        for(int i=0;i<n;i++){
            parent[i]=i;
        }
    }

    int findparent(int node){
        if(parent[node]==node){
            return node;
        }
    }
}
```

```

        parent[node]=findparent(parent[node]);
        return parent[node];
    }

    void unionit(int u,int v){
        int pu=findparent(u);
        int pv=findparent(v);

        if(pu==pv) return;

        if(size[pu]<size[pv]) swap(pu,pv);

        parent[pv]=pu;
        size[pu]+=size[pv];
    }
};

```

2. Solution Implementation

```

class Solution {
public:
    int countCompleteComponents(int n, vector<vector<int>>& edges) {
        // basically dsu se krna he to
        // edges=(k*k-1)/2
        // k nikalne ke liye number of nodes chahiye in a component
        // aur curren edge count chahiye
        dsu ds(n);

        for(auto &it:edges){
            int u=it[0];
            int v=it[1];
            ds.unionit(u,v);
        }

        unordered_map<int,int> nodes;
        unordered_map<int,int> edgecnt;

        for(int i=0;i<n;i++){ // node nikaalte waqt saare node consider kreng
            int boss=ds.findparent(i);
            nodes[boss]++;
        }
    }
};

```

```

    for(auto &it:edges){ // edge nikaalte waqt sirf u consider krenge jo
        int u=it[0];
        int boss=ds.findparent(u);
        edgecnt[boss]++;
    }

    int ans=0;

    for(auto &it:nodes){
        int boss=it.first;
        int k=it.second;
        int ecnt=edgecnt[boss];

        if(ecnt==(k*(k-1))/2) ans++;
    }

    return ans;
}
};

```